

Sr.	Unit No.	Question	BL	CO
1	1	Define Operating System and discuss functions of Operating System.	U	C01
2	1	Describe features of Operating System in brief.	U	C01
3	1	Explain Batch Operating System with suitable example.	U	C01
4	1	Describe Multiprogramming Operating System with suitable example.	U	C01
5	1	Discuss Time-Sharing Operating System with proper example.	U	C01
6	1	Write a short note on Single User and Multiuser Operating System.	R	C01
7	1	Discuss services of Operating System.	U	C01
8	1	What is System Call? Explain types of System Calls.	U	C01
9	1	Explain System Program with its types.	U	C01
10	1	Describe different types of Operating System Structures with proper diagram.	U	C01
11	2	Define Process. State the difference between Process and Program.	R	C02
12	2	Explain Process Life-Cycle with a suitable diagram.	U	C02
13	2	Explain Process States Transitions with a suitable diagram.	U	C02
14	2	Write a short note on PCB (Process Control Block).	R	C02
15	2	Discuss Context Switching with proper example.	U	C02
16	2	What is Thread? State the difference between Thread and Process.	R	C02
17	2	Explain Single VS Multiple Threaded Process.	U	C02
18	2	Explain types of Thread with suitable example. OR Explain User Level Thread and Kernel Level Thread."	U	C02
19	2	Define the following terms: i. Scheduling ii. Scheduler iii. Scheduling Algorithm	R	C02
20	2	Explain different types of Schedulers in brief. OR Explain Short, Medium- and Long-Term Scheduler in brief.	U	C02
21	2	Discuss Scheduling Criteria.	U	C02
22	2	Explain FCFS (First Come, First Served) Scheduling Algorithm with a suitable example.	U	C02
23	2	Explain SJF (Shortest Job First) Scheduling Algorithm with a suitable example.	U	C02
24	2	Explain SRTN (Shortest Remaining Time Next) Scheduling Algorithm with a suitable example.	U	C02
25	2	Explain RR (Round Robin) Scheduling Algorithm with a suitable example.	U	C02
26	2	Explain Preemptive Priority Scheduling Algorithm with a suitable example.	U	C02
27	2	Explain Non-Preemptive Priority Scheduling Algorithm with a suitable example.	U	C02
28	2	Consider four processes: P1, P2, P3, and P4, with an arrival time of 0 ms for all. Their respective burst times are 6 ms, 8 ms, 7 ms, and 3 ms. Draw the Gantt chart and calculate the Finish Time, Average Waiting Time and Turn Around Time using FCFS scheduling algorithm.	A	C02
29	2	Consider four processes: P1, P2, P3, and P4, with an arrival time of 0 ms for all. Their respective burst times are 6 ms, 8 ms, 7 ms, and 3 ms. Draw the Gantt chart and calculate the Finish Time, Average Waiting Time and Turn Around Time using SJF scheduling algorithm.	A	C02

30	2	Consider four processes: P1, P2, P3, and P4, with an arrival time of 0 ms, 1 ms, 3 ms and 5 ms respectively. Their respective burst times are 6 ms, 4 ms, 2 ms, and 1 ms. calculate the Finish Time, Average Waiting Time and Turn Around Time using SRTN scheduling?	A	C02
31	2	Consider four processes: P0, P1, P2, and P3, with an arrival time of 0 ms, 1 ms, 3 ms and 5 ms respectively. Their respective burst times are 6 ms, 4 ms, 2 ms, and 1 ms. Draw the Gantt chart and calculate the Finish Time, Average Waiting Time and Turn Around Time using Round Robin scheduling?	A	C02
32	3	Define Inter-Process Communication(IPC). Explain independent and cooperating process with suitable example.	U	C03
33	3	Why we are required process cooperation? List out the reason required for process cooperation.	R	C03
34	3	Discuss models for Inter-Process Communication with suitable diagram. OR Describe Message Passing and Shared Memory Model for IPC.	U	C03
35	3	What is Race Condition? Explain with suitable example.	U	C03
36	3	What is Critical Section and Mutual Exclusion? Explain with proper example.	U	C03
37	3	Define following terms: I. Bounded Waiting II. Progress	R	C03
38	3	List out the mechanisms for achieving mutual exclusion with busy waiting and explain Disabling Interrupts and Shared Lock Variable with a suitable example.	U	C03
39	3	List out the mechanisms for achieving mutual exclusion with busy waiting and explain Strict Alteration and Test and Set Lock (TSL) Instruction with a suitable example.	U	C03
40	3	What is Deadlock? Explain characteristics of deadlock.	U	C03
41	3	Explain Deadlock Ignorance with suitable example.	U	C03
42	3	Discuss Deadlock Detection and Recovery with suitable example.	U	C03
43	3	Describe deadlock Avoidance and Prevention with suitable example.	U	C03
44	3	What is Memory? Explain types of memory in brief.	U	C03
45	3	State the difference between Logical Address and Physical Address.	R	C03
46	4	Explain fixed (static) and variable (dynamic) partition memory allocation technique with a suitable example.	U	C04
47	4	Describe swapping technique with suitable example.	U	C04
48	4	Discuss Non-Contiguous memory allocation with simple example.	U	C04
49	4	Explain Paging technique with a suitable example.	U	C04
50	4	Explain Segmentation technique with a suitable example.	U	C04
51	4	Explain memory allocation strategies with a proper example. OR Explain First Fit, Next Fit, Best Fit and Worst Fit with a suitable example.	U	C04
52	4	Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB . These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. Perform the allocation of processes using first fit, next fit, best fit, and worst fit memory allocation scheme.	A	C04
53	4	Suppose we have some memory frames of 16KB, 14KB, 4KB, 20KB, 18KB, 7KB, 9KB, 12KB, and 15K . Similarly, we, have pages of 12KB, 10KB, and 9KB .	A	C04

		How you will allocate the memory if you follow first fit, next fit, best fit , and worst fit memory allocation scheme.		
54	4	Explain FIFO page replacement algorithm with suitable example.	U	C04
55	4	Explain LRU page replacement algorithm with suitable example.	U	C04
56	4	Explain Optimal page replacement algorithm with suitable example.	U	C04
57	4	Consider page reference string 1, 3, 0, 3, 5, 6, 3 with 3-page frames. Calculate the number of page faults using FIFO Page Replacement Algorithm.	A	C04
58	4	Consider Page Reference String 3, 1, 2, 1, 4, 5, 4, 3, 2 with 3-page frames. Calculate the number of page faults using LRU page replacement algorithm.	A	C04
59	4	Consider the page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3 with 4-page frames. Calculate the number of page faults using LRU Page Replacement Algorithm.	A	C04
60	4	Consider the page references 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3 with 4-page frame. Calculate the number of page fault using Optimal Page Replacement Algorithm.	A	C04
61	5	What is I/O device? Explain types of devices in OS. OR What is I/O device? Explain dedicated, shared and virtual device with suitable example.	U	C05
62	5	What is I/O buffering? Explain types of I/O buffering in brief.	U	C05
63	5	Draw neat and clean diagram of internal structure of Hard Disk Drive.	U	C05
64	5	Define the following terms: i. Seek Time ii. Rotational Latency iii. Access Time	R	C05
65	5	Explain First Come First Serve (FCFS) disk scheduling algorithm with suitable example.	U	C05
66	5	Explain Shortest Seek Time First (SSTF) disk scheduling algorithm with suitable example.	U	C05
67	5	Explain SCAN and C-SCAN disk scheduling algorithms with suitable examples.	U	C05
68	5	Explain LOOK and C-LOOK disk scheduling algorithms with suitable examples.	U	C05
69	5	Define the file. Discuss different types of files in brief.	U	C05
70	5	List operations perform on files and discuss all in brief.	U	C05
71	5	List attributes of file and explain in brief.	U	C05
72	5	Describe file access methods with a suitable example. OR Describe Sequential, Direct and Indexed file access methods with a suitable example.	U	C05
73	5	Define the directory. Write a short note on Directory Structures in OS.	R	C05